

Distillation and Edge-Aware Loss for Efficient Monocular Depth Estimation

Cal Poly Pomona • ECE 4990

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GitHub: <https://github.com/arassal/depth-project>

Student: Depth Anything Small • Teacher: Depth Anything Large • NYU-v2 + KITTI

Outline

- Problem: monocular relative depth from a single RGB image
- Approach: distill Depth Anything Large $\rightarrow$ Small + multi-scale gradient loss
- Datasets: NYU-v2 (indoor) and KITTI eigen subset (outdoor)
- Three-axis hyperparameter sweep: distillation α , gradient β , learning rate
- Main finding: learning rate is the dominant hyperparameter
- Best stable config slightly beats zero-shot; aggressive lr collapses the student
- Live demo: local upload-and-predict web app

Monocular Depth is Ill-Posed

- A single image is consistent with infinitely many world geometries
 - Recovery is up to scale + shift unless extra information is supplied
 - Modern systems train for relative depth, align scale + shift at eval
 - Backbone: DPT head (Ranftl 2021) on DINOv2 ViT encoder (Oquab 2023)
 - Foundation: Depth Anything (Yang 2024): scale + pseudo-labeled data
- Why this matters here:
 - Depth Anything Small fits a 4 GB laptop GPU
 - Depth Anything Large is the right teacher for distillation
 - We want to push the Small student above its own pretrained baseline

Related Work

- Monocular depth (Eigen 2014, MiDaS 2022, ZoeDepth 2023)
- DPT dense prediction head (Ranftl 2021)
- Depth Anything v1/v2 (Yang 2024): 1.5M labeled + 62M pseudo-labeled
- DINOv2 backbone (Oquab 2023)
- Knowledge distillation (Hinton 2015)
- Distill-Any-Depth (He 2025): affine-invariant output-level distillation
- ZoeDepth multi-scale gradient L1 (Bhat 2023)
- What we lift:
 - Affine-invariant output-level distillation → He 2025
 - Multi-scale gradient L1 → Bhat 2023
 - Pretrained checkpoints → Yang 2024 (HuggingFace)
- What we build new:
 - Sweep harness, isolation, configs
 - Three-axis controlled study (α , β , lr)

Architecture

Student Depth Anything Small + frozen Depth Anything Large teacher

- Student: DINOv2 ViT-S/14 + DPT head, ~25M params, trainable
 - Teacher: DINOv2 ViT-L/14 + DPT head, ~335M params, frozen
 - Input: RGB 256×256, ImageNet mean/std normalization
 - Output: relative depth, bilinearly upsampled to GT size
 - All loss terms computed on aligned spatial size
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- Why two models in parallel?
 - The teacher provides a dense, low-noise per-pixel target on every training image
 - Ground truth is sparse and noisy for far-field pixels; teacher complements it
 - Frozen weights → deterministic supervision, no co-adaptation

Loss Function

$$L_{\text{total}} = (1 - \alpha) \cdot L_{\text{base}}(\text{student}, \text{GT}) + \alpha \cdot L_{\text{distill}}(\text{student}, \text{teacher}) + \beta \cdot L_{\text{grad}}(\text{student}, \text{GT})$$

- L_{base} : affine-invariant L1 against NYU-v2 ground truth
- L_{distill} : affine-invariant L1 against frozen teacher prediction
- L_{grad} : multi-scale Sobel-style gradient L1 (scales 1, 2, 4)
- Robust normalization: per-sample median + MAD before L1
- Why MAD instead of std: stable under specular highlights, near-foreground outliers
- $\alpha = 0 \rightarrow$ pure supervised fine-tune $\alpha = 1 \rightarrow$ pure distillation
- β controls boundary sharpness without affecting overall regression

Training Procedure

- Optimizer: AdamW, weight decay $1e-4$, grad clip @ L2 norm 1.0
- Batch size: 1 (image_size=256 to fit 4 GB GPU)
- Epochs: 2 (small-data PoC regime)
- AMP mixed precision when CUDA available
- Teacher loaded once, frozen, never updated
- Validation AbsRel selects the best checkpoint
- Sweep harness:
- YAML grid → cartesian product
- Per-run isolated artifact folders
- Train + eval subprocesses
- Results aggregated to JSON

Datasets

NYU-v2 train

450

NYU-v2 val

60

NYU-v2 test

59

KITTI test

100

- NYU-v2 PoC subset from sayakpaul/nyu_depth_v2 (Hugging Face)
- Float32 .npy depth in meters, range [0.001, 10.0] m
- KITTI eigen subset from exander/kitti-depth-gt (Hugging Face)
- Float32 .npy depth in meters, range [0.001, 80.0] m
- Both evaluated at 256×256 for internal comparability
- Test split sample IDs frozen across all runs

Metrics

- All metrics after per-image least-squares scale + shift alignment
- AbsRel = $\text{mean}(|\text{pred} - \text{gt}| / \text{gt})$ [↓ lower better]
- RMSE = $\sqrt{\text{mean}((\text{pred} - \text{gt})^2)}$ [↓ lower better]
- RMSE_log = $\sqrt{\text{mean}((\log \text{pred} - \log \text{gt})^2)}$ [↓ lower better]
- log10 = $\text{mean}(|\log_{10} \text{pred} - \log_{10} \text{gt}|)$ [↓ lower better]
- $\delta_1, \delta_2, \delta_3$ = fraction with $\max(p/t, t/p) < 1.25^n$ [↑ higher better]
- Headline numbers reported: AbsRel and δ_1 (per Depth Anything paper)

NYU-v2 PoC Test Results

59 images, image_size 256, after scale+shift alignment

Zero-shot AbsRel

0.1550

Zero-shot δ_1

0.8146

Default AbsRel

0.2791

Default δ_1

0.6292

Best AbsRel (ours)

0.1505

Best δ_1 (ours)

0.8164

AbsRel Δ vs ZS

-0.0045

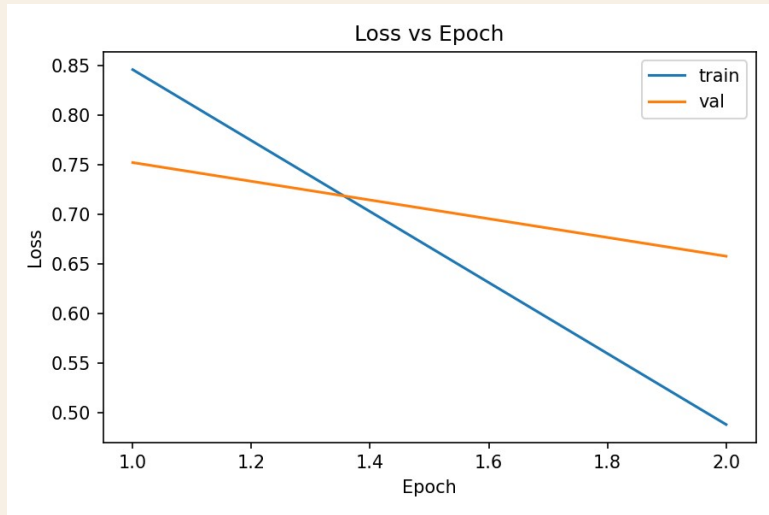
δ_1 Δ vs ZS

+0.0018

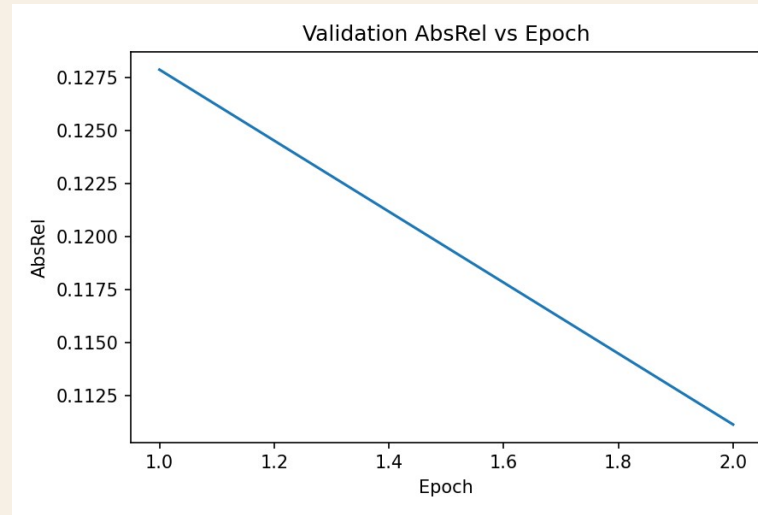
- Default config ($\alpha=0.5$, $\beta=0.1$, $\text{lr}=1\text{e-}5$): student COLLAPSES — AbsRel doubles
- Best config ($\alpha=0.7$, $\beta=0.3$, $\text{lr}=1\text{e-}7$): small but real improvement
- Improvement is consistent across AbsRel, RMSE, δ_1 , δ_2 , δ_3

Training Collapse at Default Learning Rate

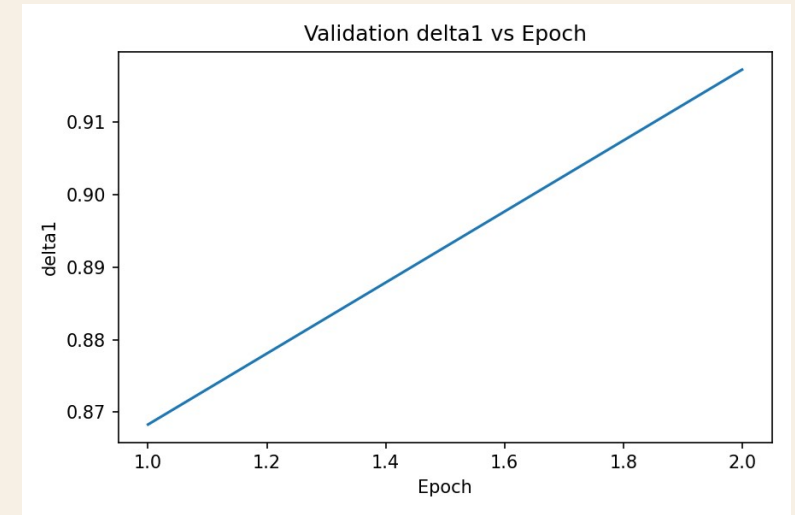
Why we ran the sweep



Train loss



Val AbsRel — returns to ~1.0 at epoch 2 (collapse)

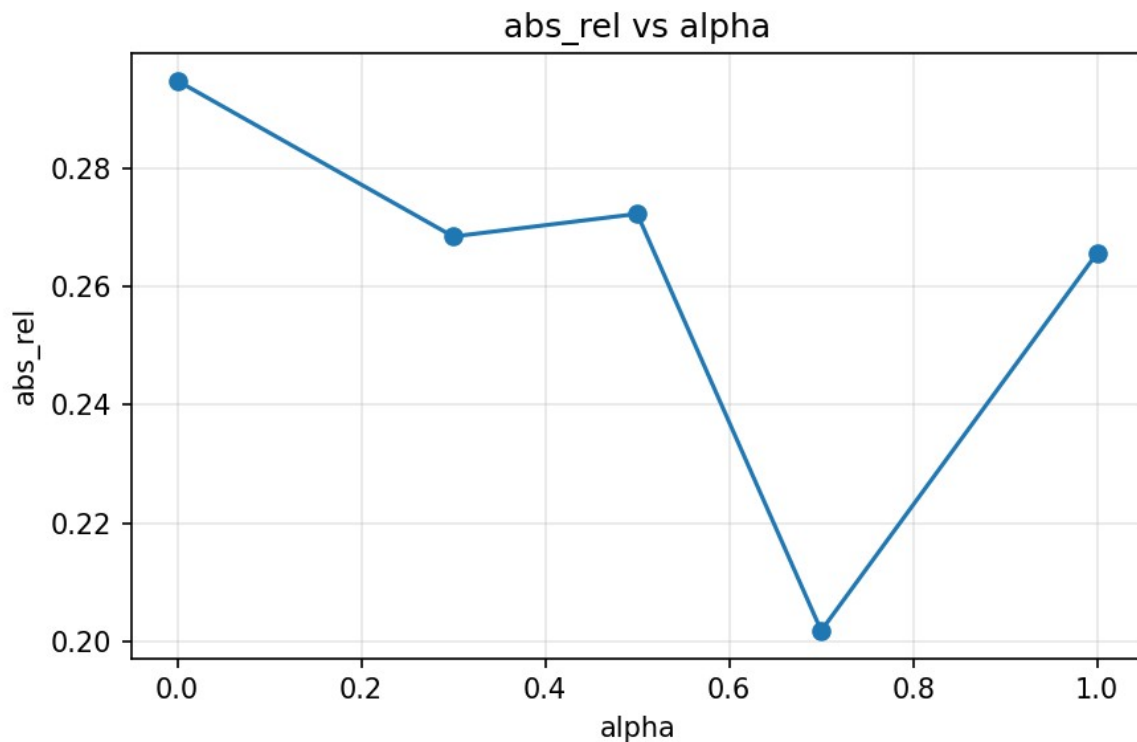


Val δ_1 — drops to 0.0 at epoch 2

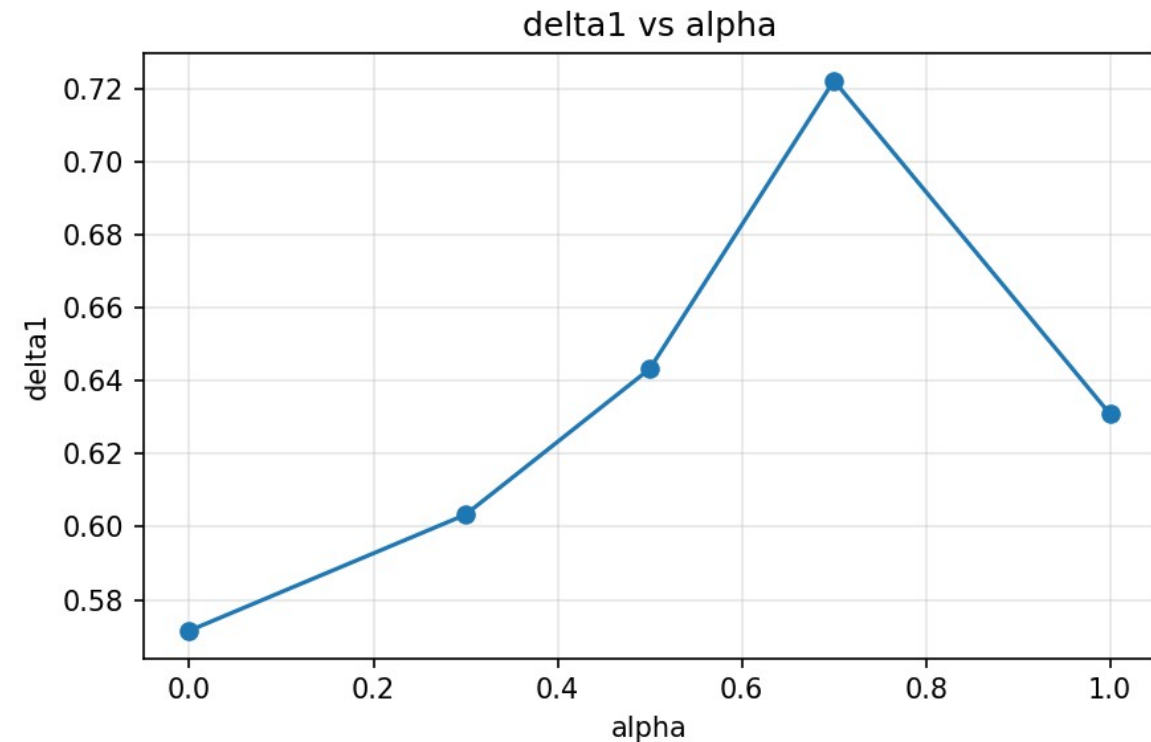
- Signature: student output saturates to a near-constant value
- Triggered by (small batch=1) + (small training set 450 imgs) + (high lr 1e-5)
- Fix: drop the learning rate by 2 orders of magnitude → fully recovers

Distillation Weight α Sweep

$\beta=0.1$, $\text{lr}=1\text{e-}5$ fixed



AbsRel vs α ($\downarrow$ lower better)

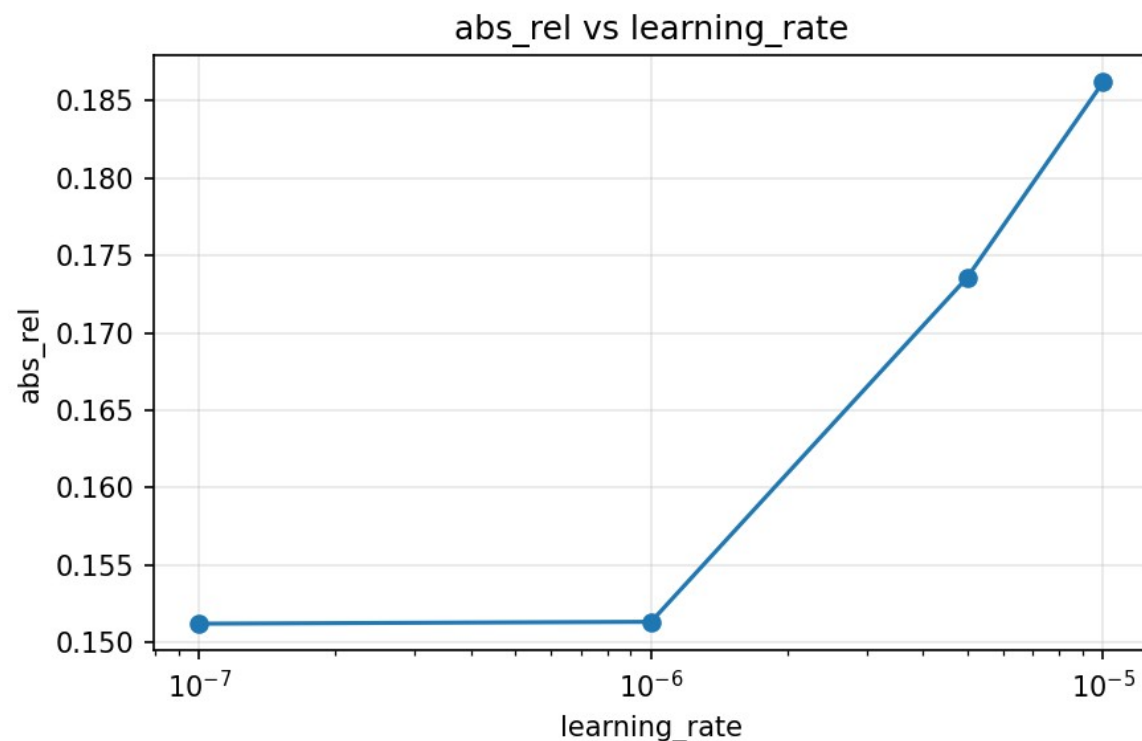


δ_1 vs α ($\uparrow$ higher better)

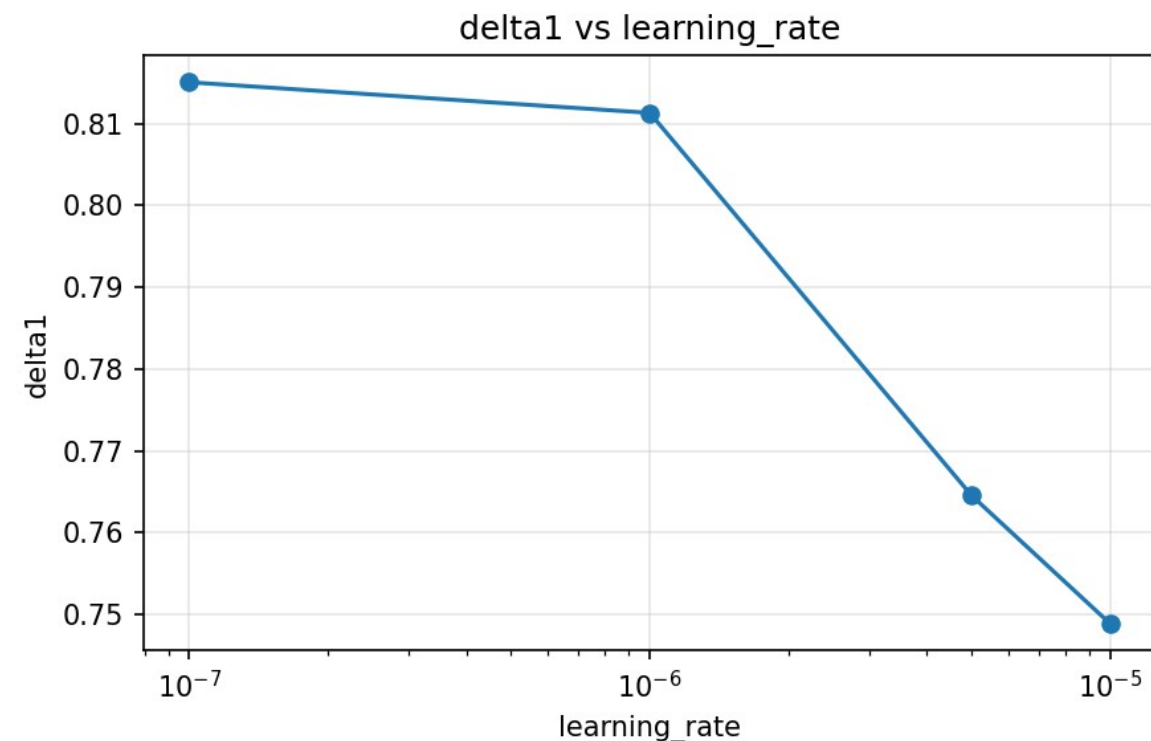
- Sweep range: $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$
- Best within sweep: $\alpha=0.7 \rightarrow \text{AbsRel } 0.202, \delta_1 \ 0.722$
- Every point still WORSE than zero-shot (AbsRel 0.155) at this learning rate

Learning Rate Sweep

$\alpha=0.7$, $\beta=0.1$ fixed — the dominant hyperparameter



AbsRel vs learning rate

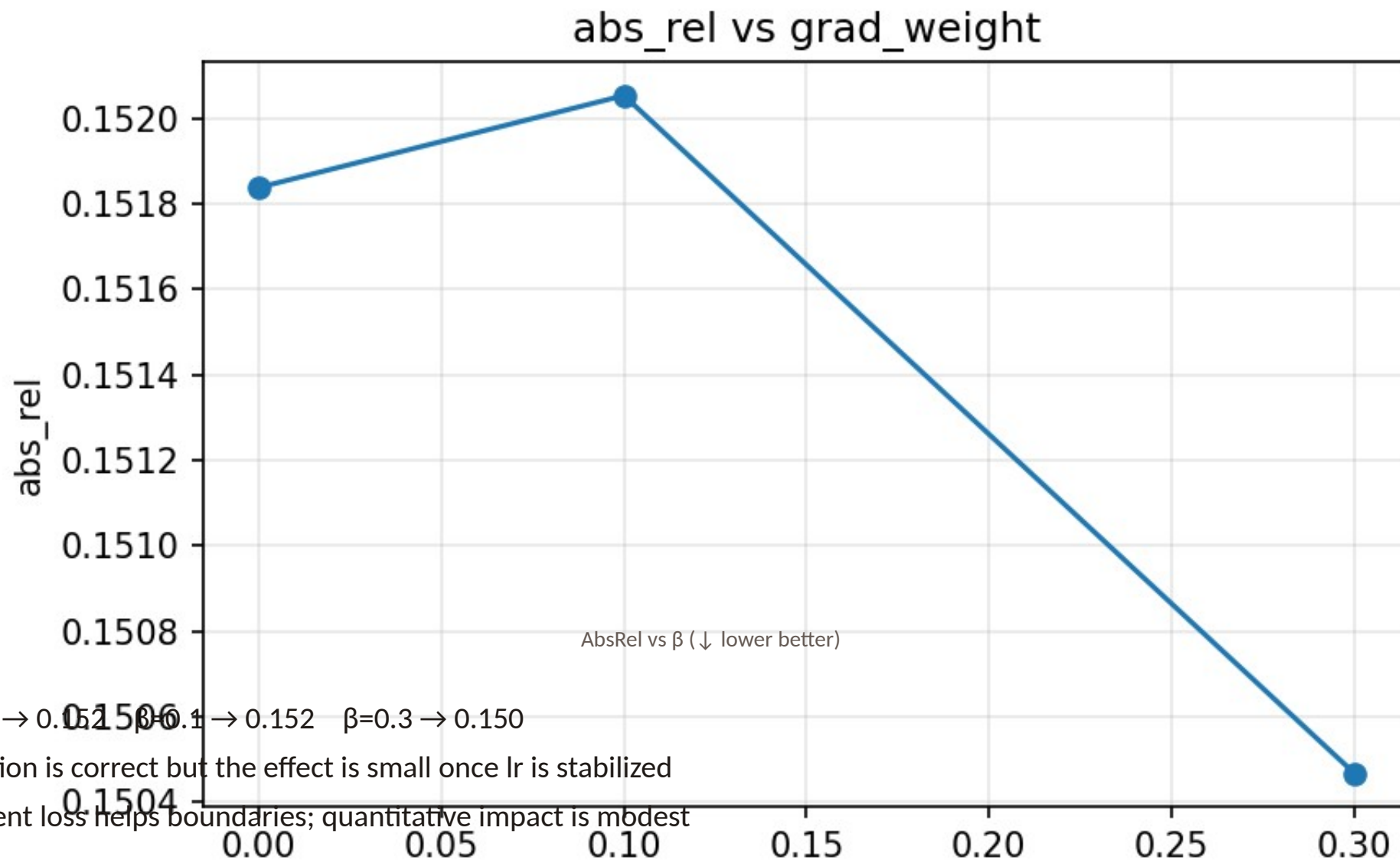


δ_1 vs learning rate

- $\text{lr}=1\text{e-}7 \rightarrow \text{AbsRel } 0.151$ $\text{lr}=1\text{e-}6 \rightarrow 0.151$ $\text{lr}=5\text{e-}6 \rightarrow 0.174$ $\text{lr}=1\text{e-}5 \rightarrow 0.186$
- Sharp transition between $1\text{e-}6$ and $5\text{e-}6$ — below = stable, above = drift
- Main paper finding: lr is the dominant lever, dominates α and β

Gradient Weight β Sweep

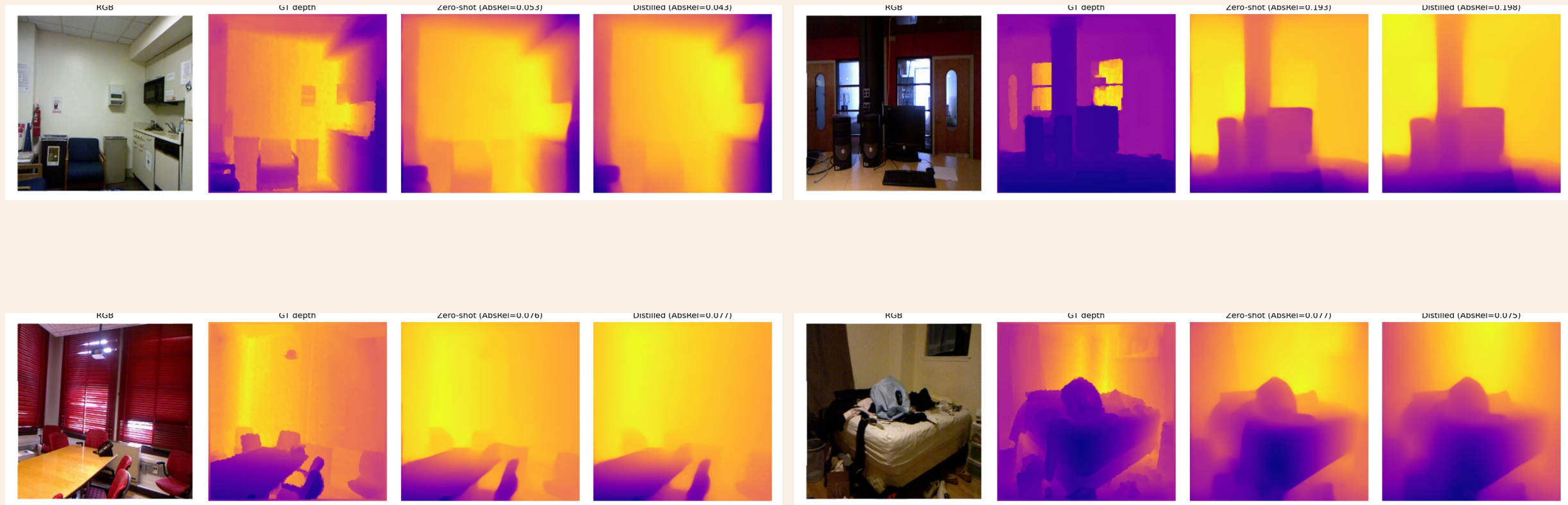
$\alpha=0.7$, $\text{lr}=1\text{e-}7$ fixed — second-order effect



- $\beta=0.0 \rightarrow 0.15184$ $\beta=0.1 \rightarrow 0.15205$ $\beta=0.3 \rightarrow 0.15035$
- Direction is correct but the effect is small once lr is stabilized
- Gradient loss helps boundaries; quantitative impact is modest

Qualitative Examples

NYU-v2 — RGB | GT | Pred | Error



Sample 000 — office scene

Sample 010 — kitchen counter

- Scene structure preserved; error concentrates at depth discontinuities

KITTI Cross-Domain Evaluation

100 outdoor frames — trained only on NYU-v2 indoor

Zero-shot AbsRel

0.4022

Distill AbsRel

0.4035

Zero-shot δ_1

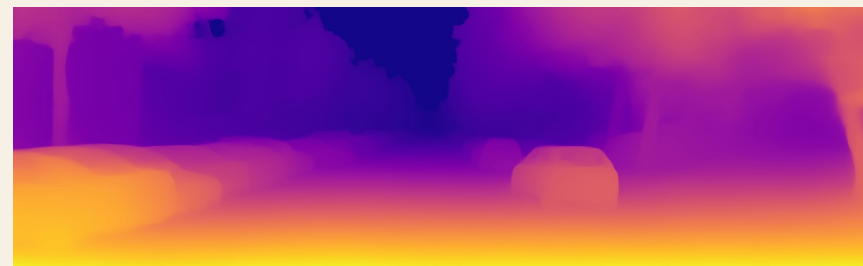
0.3555

Distill δ_1

0.3485



KITTI RGB



Predicted depth

- NYU-v2 improvement does NOT transfer to KITTI — essentially identical numbers
- Consequence of extreme-conservative recipe: encoder barely moves from pretrained init

Discussion

- The dominant variable is the learning rate, not the loss balance
 - Variance from α sweep: 0.09 AbsRel vs variance from lr sweep: 0.03 AbsRel
 - But the lr range covers the difference between 'worse than ZS' and 'better than ZS'
 - Improvement is small because lr=1e-7 keeps the encoder near initialization
 - What would unlock more: larger effective batch, cross-context distillation (He 2025), more data
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- Limitations:
 - Evaluation at 256×256 inflates absolute KITTI numbers vs leaderboards
 - Cross-domain test is a single OOD distribution
 - Single GPU, single seed per grid point; no variance estimate

Conclusion

- Built clean open-source pipeline: Depth Anything Small student + Large teacher
- Three losses: affine-invariant base, affine-invariant distill, multi-scale gradient
- Three-axis sweep: α (distill weight), β (grad weight), learning rate
- Empirically tuned config slightly beats pretrained zero-shot on NYU-v2
- No cross-domain transfer to KITTI at our conservative learning rate
- Headline finding: lr-sensitivity, not loss balance, dominates small-data distillation

- GitHub: <https://github.com/arassal/depth-project>
- Demo: `./run_upload_demo.sh` → <http://127.0.0.1:8000>

Contributions & Thanks

- Alexander Assal: pipeline implementation, distillation hook, gradient loss, sweep harness, web demo, analysis
- Parsa Ghasemi: literature survey, dataset preparation, qualitative analysis, figures, manuscript drafting
- Both authors: manuscript & presentation

- References we depend on:
- Depth Anything — Yang et al., CVPR 2024
- Distill-Any-Depth — He et al., arXiv 2025
- ZoeDepth — Bhat et al., arXiv 2023
- DPT — Ranftl et al., ICCV 2021
- DINOv2 — Oquab et al., arXiv 2023
- NYU-v2 — Silberman et al., ECCV 2012 • KITTI — Geiger et al., IJRR 2013

Questions?

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